

Lab: JavaScript Fundamentals (Variables, Methods, Conditions, Strings, and Dates)

Course: Web Design

Instructor: Mohammad Zafar Shafaq

Introduction

In this lab, students will apply the fundamental concepts of JavaScript programming. They will work with variables, operators, conditional statements, string methods, and date methods to create interactive programs.

The purpose of this lab is to strengthen logical thinking and problem-solving skills while preparing students for more advanced JavaScript topics.

Task 1: Student Information System

Requirements

Create variables for:

- Student Name
- Student ID
- Department
- GPA

Display the information using:

`document.write()`

Challenge

Display a message:

- "Excellent Student" if $GPA \geq 3.5$
- "Good Student" if $GPA \geq 2.5$
- "Needs Improvement" otherwise

Task 2: Age Verification System

Requirements:

Ask the user to enter their age using:

`prompt()`

Use conditional statements to determine:

- Under 18 → Minor
- 18–60 → Adult
- Above 60 → Senior Citizen

Display the result.

Task 3: String Processing Tool

Requirements

Ask the user to enter their full name.

Perform the following operations:

- Convert to Uppercase
- Convert to Lowercase
- Count total characters
- Display first character
- Display last character

Example Methods

`toUpperCase()`

`toLowerCase()`

`length`

`charAt()`

Task 4: Email Validator

Requirements

Create a navigation menu using:

Ask the user to enter an email address.

Check whether:

- It contains "@"
- It contains ".com"

Display:

- Valid Email
- Invalid Email

Task 5: Date Information System

Requirements

Display:

- Current Date
- Current Day
- Current Month
- Current Year

Challenge Task:

Student Scholarship Eligibility System

Create a program that asks the user:

- Name
- GPA
- Age

Conditions:

Eligible for Scholarship

- $\text{GPA} \geq 3.5$
- $\text{Age} \leq 25$

Partially Eligible

- $\text{GPA} \geq 3.0$

Not Eligible

- Otherwise

Display a complete report.

Submission:

Students should show the progress of work on the day of LAB with IS department LAB Technician, before 3:00 PM. The complete project will be checked in coming class session on next week.